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CRITICAL THINKING FRAMEWORK

The Houses of Thought Framework

A structured method for reasoning clearly and building stronger conclusions.

Based on John Trapasso's model, derived from the Paul-Elder Framework for Critical Thinking.

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SECTION 1

Why Structured Thinking Matters

Most arguments fail not because people are unintelligent, but because their reasoning is unstructured.

Every day, people make decisions, form opinions, and draw conclusions. But how many of those conclusions are actually well-supported? Many arguments fail because people:

- Jump to conclusions without examining evidence
- Ignore perspectives that challenge their beliefs
- Rely on weak or unexamined assumptions
- Fail to ask the right questions

The Houses of Thought framework solves this by providing a clear, step-by-step structure for reasoning. Think of it like building a house: **a strong house requires a strong foundation**.

If your foundation — your facts and evidence — is weak, everything built on top of it will be unstable. If you skip steps, your conclusions will have gaps. The Houses of Thought ensures that every conclusion is traceable back through logical steps to verified information.

SECTION 2

The Full Reasoning Flow

The complete structure of the Houses of Thought, from purpose to consequences.
Click any layer to learn more.

Just as an architect designs a building from the ground up, critical thinkers must build their reasoning from the bottom up:

- 8 Implications & Consequences >
- 7.2 Overarching Conclusion >
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- 5 Information / Facts >
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Reasoning flows in two directions: **top-down** when formulating questions and perspectives, and **bottom-up** when building from evidence toward conclusions. The process is cyclical — new evidence may require revisiting earlier steps.

SECTION 3

Concepts

The key terms and definitions that establish shared understanding.

Before defining your purpose or asking any questions, it's important to identify the **key concepts** that will underpin your house. Concepts are the foundational terms and definitions that everyone involved must agree on to reason effectively.

Without clearly defined concepts, two people can use the same word to mean entirely different things — leading to confusion and flawed conclusions.

EXAMPLE

Topic: "Should salary caps exist in professional sports?"

Key Definition: "Salary cap" — a league-imposed limit on the total amount a team can spend on player salaries.

Key Definition: "Competitive balance" — a state where teams have roughly equal chances of winning, often measured by win distribution across a league.

Clearly defining your concepts early prevents misunderstandings and ensures that your reasoning is built on a solid intellectual foundation. If two people define "competitive balance" differently, they may reach entirely different conclusions from the same evidence.

SECTION 4

Purpose

Every reasoning project begins with a clear purpose.

Before asking any questions, you must understand *why* you are reasoning in the first place. Purpose answers the question: **"Why are we analyzing this?"**

A clear purpose keeps your reasoning focused and prevents you from drifting into unrelated areas. Without purpose, your house becomes aimless.

EXAMPLE PURPOSES

- Understanding a complex social problem
- Making an informed decision about a career change
- Evaluating whether a public policy is effective
- Exploring the ethical implications of a new technology

Your purpose doesn't need to be complicated. It simply needs to be clear enough that anyone reading your house understands what you're trying to accomplish.

SECTION 5

Overarching Question

Reasoning begins with a clear, central question.

The overarching question is the single most important element of your house. It defines what you are trying to answer. Everything in the Houses of Thought — every sub-question, every piece of evidence, every conclusion — must connect back to this question.

A good overarching question is specific enough to be answerable but broad enough to require structured reasoning.

EXAMPLES

"Should salary caps exist in professional sports?" — Requires examining economics, fairness, competitive balance, and player rights.

"Has artificial intelligence done more good than harm?" — Requires examining technology, ethics, economics, and society.

A vague or poorly defined question leads to vague reasoning. Spend time crafting your overarching question carefully — it is the compass for your entire house.

SECTION 6

Sub-Questions

Large questions must be broken into smaller, manageable pieces.

No complex question can be answered in a single step. Sub-questions break the overarching question into focused, answerable parts. Each sub-question is examined from a specific point of view, ensuring comprehensive coverage of the issue.

In complex situations, there can be hundreds or even thousands of sub-questions. The more thorough your sub-questions, the stronger your final conclusion will be.

EXAMPLE

Main Question: "Should salary caps exist in professional sports?"

- How do salary caps affect competitive balance between teams?
- How do salary caps affect player income and career freedom?
- How do salary caps affect fan engagement and viewership?
- What are the economic effects on team owners and leagues?
- How do salary caps compare to free-market alternatives?

Each sub-question will go through its own process of gathering information, examining assumptions, and reaching a sub-conclusion before contributing to the final answer.

SECTION 7

Points of View of Sub-Questions

Good reasoning requires examining each sub-question from multiple perspectives.

One of the most common reasoning failures is seeing an issue from only one perspective. The Houses of Thought addresses this by requiring reasoning from multiple points of view. Each sub-question should be examined from these different lenses:

Individual
Any individual's point of view — a specific person's perspective shaped by their unique experiences, beliefs, and circumstances.
e.g. A player, a fan, a team owner, a coach, a taxpayer

Group POVs
Perspectives from communities, organizations, demographics, or societal groups.
e.g. Students, teachers, governments, businesses, patients, athletes

Ideas / Disciplines
Academic disciplines, theoretical frameworks, or schools of thought.
e.g. Science, economics, ethics, technology, psychology, law

Each point of view produces different questions, highlights different evidence, and may lead to different conclusions. By systematically examining multiple POVs for each sub-question, you build a more complete and balanced understanding of the issue.

SECTION 8

Personal Foundational Point of View

Understanding the lenses through which you naturally see the world.

Before examining any issue, it's critical to understand **your own foundational perspective** — the deeply ingrained lenses through which you naturally interpret the world. These personal perspectives shape how you ask questions, what evidence you notice, and how you interpret information, often without you even realizing it.

Your personal foundational point of view is made up of several dimensions:

Biological
Your physical characteristics, health, age, and neurological makeup that influence how you experience and process the world.

Social
Your social identity, cultural background, socioeconomic status, education, and the communities you belong to.

Familial
Your family structure, upbringing, values passed down through generations, and the norms you were raised with.

Individual
Your unique personality traits, personal experiences, beliefs, values, and the specific life path that makes you who you are.

Recognizing your personal foundational point of view doesn't mean eliminating bias — that's impossible. Instead, it means becoming **aware of your biases** so you can account for them in your reasoning. A person raised in a competitive sports culture may have very different instincts about salary caps than someone who values economic equality above all else.

By explicitly identifying your foundational perspective, you can better understand why you're drawn to certain conclusions and actively seek out perspectives that challenge your default assumptions.

SECTION 9

Information / Facts (The Foundation)

The foundation of all reasoning is verified, reliable information.

Information is the bedrock of the Houses of Thought. Without solid facts, everything built on top — assumptions, inferences, conclusions — is unstable. Information should be:

- Observable — Can be seen, measured, or documented
- Measurable — Can be quantified or compared
- Supported by evidence — Backed by research, data, or credible sources
- Verifiable — Can be confirmed by others through independent sources or instruments

Critical thinkers constantly ask: *"Is this actually true? What evidence supports it? Are there facts I'm missing?"*

Strong reasoning requires strong evidence. If your facts are wrong, your conclusions will be wrong — no matter how logical your reasoning is.

SECTION 10

Assumptions

The beliefs that connect facts to conclusions.

Assumptions are beliefs, premises, or frameworks that help us interpret information and draw conclusions. They are the bridge between "what we know" and "what we conclude." Every argument relies on assumptions — the question is whether those assumptions are justified.

EXAMPLE

Fact: Salary caps limit how much teams can spend on players.

Assumption: Limiting spending creates competitive balance by preventing wealthy teams from buying all the best players.

This assumption may seem reasonable, but it should be examined: *Do salary caps actually create competitive balance? Or do wealthy teams find ways around them?*

The Houses of Thought framework categorizes assumptions into three types:

- **Unknown Unknowns:** Things you don't know that you don't know — blind spots in your reasoning
- **Foundational Assumptions:** Underlying beliefs or premises taken for granted that shape your reasoning
- **Concepts that Shape Inferences:** Specific pieces of evidence paired with the inference or logical leap drawn from them — these bridge observable facts to broader conclusions

SECTION 11

Unknown Unknowns

The factors you couldn't predict.

Sometimes, factors appear that could not have been anticipated. These are called "unknown unknowns" — things you didn't know that you didn't know. They can fundamentally change the landscape of an issue.

EXAMPLES OF UNKNOWN UNKNOWNS

- A sudden technological breakthrough that changes an entire industry
- New government regulations that shift the rules of the game
- Unexpected scientific discoveries that overturn established knowledge
- Unforeseen events (pandemics, natural disasters, economic crises)

Critical thinkers must remain flexible and humble. No reasoning framework can predict everything. The Houses of Thought helps by making your reasoning transparent — so when unknown unknowns appear, you can quickly identify which parts of your house are affected and need revision.

SECTION 12

Logical Inference

The logical step between information and conclusions.

Inference is the reasoning engine of the Houses of Thought. It is the process of drawing conclusions from available information and assumptions. Think of it as the logical leap from "what I know" to "what I conclude."

Facts
Provide the foundation
↓

Assumptions
Shape interpretation
↓

Inference
Connects them logically
↓

Conclusion
The result

Every inference must be justified. A logical leap that cannot be explained or defended is a weakness in your reasoning. Critical thinkers ask: *"Does this conclusion actually follow from the evidence? Or am I making an unjustified jump?"*

SECTION 13

Sub-Conclusions

Intermediate conclusions that build toward the final answer.

Complex reasoning rarely produces a single conclusion in one step. Instead, each sub-question produces its own sub-conclusion — an intermediate answer based on the information, assumptions, and reasoning specific to that question.

EXAMPLE SUB-CONCLUSION

Sub-Question: "How do salary caps affect competitive balance?"

Sub-Conclusion: "Salary caps reduce payroll inequality between teams, but wealthy teams often circumvent them through creative contract structures, limiting their effectiveness."

Sub-conclusions are the building blocks of your final conclusion. Each one should be clearly traceable back to specific evidence and reasoning. Together, they provide the support structure for the overarching conclusion.

SECTION 14

Overarching Conclusion

The final answer to the original question.

The overarching conclusion is the roof of the Houses of Thought. It is the single, synthesized answer to your overarching question, built from all of your sub-conclusions.

A strong overarching conclusion should be supported by:

- Verified facts and reliable evidence
- Carefully examined assumptions
- Sound logical reasoning
- Multiple sub-conclusions from different perspectives

Strong conclusions require strong foundations. If you've built each layer of the house carefully — gathered solid evidence, examined your assumptions, reasoned logically, and considered multiple perspectives — your conclusion will be defensible and well-supported.

SECTION 15

Implications vs. Consequences

Understanding the difference between predicted and actual outcomes.

Implications
Predicted outcomes that *might* happen if the conclusion is correct. These are forward-looking projections.

EXAMPLE
Throwing a ball near a window **could** break it.

Consequences
Real-world results that *actually* happen. These are observed, documented outcomes.

EXAMPLE
Throwing a ball **broke** the window.

Analyzing implications helps people plan for possible outcomes before they happen. Recording consequences as they unfold allows you to compare predictions against reality and refine your reasoning.

In the Houses of Thought platform, implications can be generated by AI based on your conclusion, while consequences are entered by you as real-world events unfold.

SECTION 16

Iterative Thinking

Reasoning is a process, not a one-time event.

One of the most important principles in the Houses of Thought is that reasoning should be revisited and revised when new information appears. A conclusion reached today may need to be updated tomorrow.

If the implications are negative, unexpected, or don't match reality, thinkers should:

- Revisit and re-examine assumptions
- Gather more information and evidence
- Adjust conclusions based on new data
- Build more or stronger sub-questions
- Consider additional points of view

Reasoning is cyclical, not linear

The best thinkers are willing to change their minds when the evidence demands it. The Houses of Thought supports this by making it easy to revisit any layer of your reasoning at any time.

SECTION 17

How the Houses of Thought Platform Helps

The platform brings this framework to life with powerful tools.

Draft Full House
AI generates a complete reasoning structure — purpose, sub-questions, assumptions, and information — in one step.

Visual House Diagrams
See your reasoning as an interactive house structure. Click into any layer to explore or edit.

AI Reasoning Assistant
Generate sub-questions, test assumptions, and explore perspectives with AI guidance.

Research Mode
AI synthesizes reliable information to ground your reasoning in verified evidence.

Logic Strength Evaluation
Evaluate how well-supported your reasoning is across evidence, assumptions, and logical consistency.

Reasoning Stress Tests
Test your conclusions against counter-arguments and opposing perspectives.

The goal of the Houses of Thought platform is simple: **help you think more clearly, reason more rigorously, and build stronger conclusions**. Whether you're a student learning critical thinking, a teacher guiding reasoning projects, or a professional analyzing complex problems, the platform provides the structure and tools to do it well.

Start Building Your Houses of Thought >

Based on John Trapasso's *House of Reason* model for critical thinking, derived from the **Paul-Elder Framework** for Critical Thinking.

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Houses of Thought

An AI-powered reasoning system based on John Trapasso's model, derived from the Paul-Elder Model for critical thinking.

CREATED BY SAMIR SOGANI

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